

Designing an Intelligent Learning System For Practicing the Oboe Embouchure

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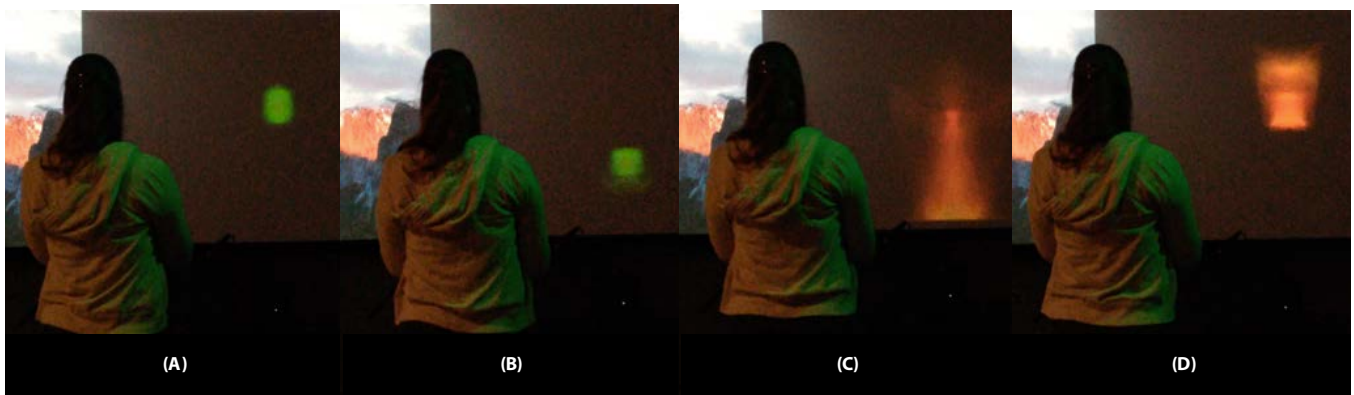


Figure 1: Our intelligent learning system. (A) Good tone (green color) with a perfect pitch (rendered in the middle) (B) Good tone with a lower pitch (C) Poor tone (reddish color) with an irregular pitch (rendered in the bottom) (D) Poor tone with a higher pitch

ABSTRACT

The oboe is considered one of the most difficult woodwind instruments to learn due to the labor involved in mastering the embouchure required to produce a steady, long tone through the instrument's reeds. Previous research in music education has suggested that using visualization or imagery as a training method, so-called audiation, helps novice learners gain a better understanding of the relationship between sound and embouchure. Inspired by this concept, we designed an interactive learning system using real-time acoustic analysis (MFCCs) and support vector machines (SVM). Our interactive visualization aims to support the user's learning process by rendering the quality of good and bad sounds as pitch indications through particle swarm animations. A pilot user study showed that this makes the learning experience more reflective and self-directed, as the learner can understand the relationship between their breathing technique and the visualization of the generated sound quality. In this paper, we intend to address issues potentially involved in designing an intelligent learning system using sensing technologies in the context of music education.

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CCS CONCEPTS

• Human-centered computing → User interface design; • Applied computing → Sound and music computing.

KEYWORDS

oboe practice, music education, intelligent tutoring system, machine learning, motivation

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1 INTRODUCTION

Although the oboe is well known for its calming, soothing sound, it is considered one of the most difficult woodwind instruments to learn. Although several factors make it difficult, such as the fingering system, tuning, and breath management, a fundamental barrier is dealing with the reed and learning the exact placement of the lips to produce the proper sound (instead of sounding like a quacking duck). It takes some time for the novice learner to produce a clear tone through the reeds, and even then, the player may have little ability to control their breathing technique and embouchure as needed to generate a steady tone with the proper pitch.

Due to the characteristics of the oboe, novice learners experience various difficulties in the early learning stages. For example, a player may not be able to play a note with the proper pitch, produce expressions, or produce a note at the intended time, negatively

affecting their timing [14]. These problems mainly occur for two reasons: (1) the non-linear nature of the double reed's physical mechanism of producing sound and (2) the embouchure, a complex neuromuscular task that requires intensive orofacial muscle activities and breathing methods to control the quality of sound through reeds. Gaunt [4] addressed a number of common difficulties in having good embouchure. These are due to a lack of (1) anatomical and physiological understanding; (2) awareness of breathing in playing; (3) connecting breathing to posture and movement; and (4) connecting breathing to music.

In order to overcome these difficulties, novice learners must take sole charge of training themselves to control the instrument and develop their musical acumen to assess their own performance through independent repetitive practice. However, this lack of supervision is considered one of the main practical reasons for the difficulty of this learning process [5], especially in the earliest days of learning. Thus, we focused on the matter of awareness of breathing in playing and connecting breathing to music in practice sessions. In the field of music education research, researchers and instructors put their best efforts into developing effective interactive teaching methods and designing intelligent learning systems to aid in unsupervised practice sessions. In the context of computer-assisted musical instrument tutoring (CAMIT), researchers have advanced the use of advanced technologies in musical instrument tutoring and learning [6, 11] through emerging sensing technologies in consumer electronics and machine learning.

Based on the concept of CAMIT, we designed an intelligent learning system using real-time acoustic analysis (mel frequency cepstral coefficients; MFCCs) and a machine learning process (support vector machines; SVM) to aid in novice oboe learners' unsupervised practice sessions. Using interactive visualization, we aimed to support the user's learning process by rendering the quality of good and bad oboe sounds as particle swarm animations through changes in green and red particles. After this visualization was developed, we designed initial user studies based on the Attention, Relevance, Confidence, and Satisfaction (ARCS) model proposed by John Keller [9] to investigate the impact of the interactive learning system and assess users' embouchure skill development. In these pilot user tests, significant changes in breathing practices were observed. Furthermore, the users' learning behaviors became more reflective and self-directed as they began to understand the relationship between their breath and the quality of the sound rendered by the visualization.

Designing an intelligent tutoring system (ITS) is a complicated task due to the various steps involved, such as conducting user research for a requirements analysis, setting up suitable sensors, collecting data, building an appropriate model, and creating an engaging user interface. The interactive system design process can be even more challenging when not all tasks are well aligned with the overarching goal of providing a desirable learning experience with user interaction, multi-modal inputs, and real-time responses. Under the theme of Delivering Sensing Technologies for Education and Learning, we revisit our design process to share the lessons we learned from a human-centered machine learning perspective addressed by [3], as well as suggestions and potential guidelines for the future design and assessment of educational systems using various sensing technologies, especially in music education.

2 DESIGN PROCESS FOR THE INTELLIGENT LEARNING SYSTEM

An ITS is a system that applies machine learning or artificial intelligence techniques to represent knowledge in complex datasets or cases for educational purposes. Such systems have substantial advantages over traditional teaching methods and rule-based training systems, as they can offer constant feedback and help each student learn technique and knowledge effectively and efficiently according to their personal learning situation. These systems can be adapted to provide personalized training through continuous monitoring and by maintaining a representation of how the student is progressing [1, 12].

However, not all ITSs are designed with engaging content and user-friendly interactions. This is why designing an ITS from a human-centered perspective is crucial. This approach allows us to recognize a human task with a goal explicitly, as well as reframe machine learning workflows based on the user context, and it enables a smoother co-adaptation of the end users and intelligent systems. It will help us incorporate the knowledge, insight, and feedback of the human (for both domain experts and end users) into the entire design process, such that hypotheses can be refined, the models can be tuned, and the interface can be created appropriately.

As shown in Figure 2, we developed a framework for an ITS that can provide high-level, user-friendly, interactive content for novice oboe learners. This framework integrates three processes: (1) an offline process for model building and mapping visual elements to inference; (2) processing the user's audio input and estimating its quality in real time; and (3) looping user behavior/actions so that our ITS can support the user's unsupervised practice sessions in addition to the traditional ITS design paradigm, which is crucial for creating intelligent systems in the music education context.

Six key design principles underpinning the development of our system were identified from user interview sessions with tutors and novice learners. As shown in Figure 2, these design principles were used to shape the user learning experience, the flexible yet robust inter/intrapersonal variations in oboe sound generation, and the approaches to each step. These principles included (1) mental models of learners (learner-centered) and enhanced traditional learning activities (not overly disruptive); (2) an engaging and interactive experience; (3) improved accessibility; (4) clear structure and navigability, so that learners can intuitively find their own insights reflected in the learning experience; (5) the capacity to work in an in-home context with consumer electronics sensors; and (6) the capacity to support and adjust its difficulty according to the learner's growth.

3 IMPLEMENTATION OF INTELLIGENT LEARNING SYSTEM

We met with 15 novice oboe learners in a summer conservatory program and collected their individual practice audio to identify features that well represented good or bad oboe sound quality. In our user observation and data collection sessions, we identified common, yet important, problems that were observed frequently, for instance, (1) poor low pitch tone, with explosive attacks and loud dynamics; (2) poor low pitch tone with shallow tone, inconsistent

accuracy and clarity; (3) unclear, and unpredictable, attack sounds; and (4) squeak notes.

We collected samples at 16-bit, 44.1 kHz, mono using a Neumann Km 84 cardioid microphone at a 5-foot distance with ambient noise. As a result, we found that there were significant differences between bad and good oboe sound samples in terms of pitch, amplitude, and length. Before we started a systemic feature engineering process, we explored potential features by visualizing good and bad samples with well-known features used for acoustic signal processing. Computational representation of such a quality difference is shown in Figure 3.

During the feature engineering process, we thoroughly analyzed the sound samples to find the most relevant features across four domains: time-related features, frequency-related features, features inspired by voice production and perception (MFCCs, LPC, PLP), and long-term features. These features were used to build a regression model that would allow us to map the accuracy to graphical elements and particle swarm animations in the visualization. MFCCs [2] are widely used in automatic speech recognition systems, which were derived from short-time spectral energies in a condensed sphere. MFCCs are derived from log-scaled mel-filterbank (MFB) energies.

An SVM algorithm was used to estimate the goodness and badness of each input sound, and the model was designed to produce inferences from 0 (perfect bad-quality sound) to 1 (perfect good-quality sound). The algorithm was applied with a Gaussian kernel training method and 32 bands of MFCC features to achieve the highest accuracy for good and bad sound estimation, as shown in Figure 4 (left). Three important features were extracted from the datasets to characterize good/bad aspects of tone—pitch, jitter, and shimmer—in addition to the MFCC features, and the results were used to create visualizations, as shown in Figure 4 (right).

Our visualization design rationale was to create interactive, intuitive animation with the concept of breathing to help the user concentrate on the air flow (embouchure) through the instrument without an additional cognitive load. We hoped they could (1) focus on the overall musical phrase; (2) direct their breath into a constant, steady air flow; (3) avoid holding their breath for a long period; and (4) integrate steady, long tones and pitches into their breathing practice. The eventual goals were to produce a good tone (or sound) and gain endurance.

4 EVALUATION OF A LEARNING EXPERIENCE USING AN ARCS MODEL

The feedback frameworks suggested in recent ITS research consider the learner an active constructor of knowledge [10]. Therefore, the most important aspect of learning with an intelligent system is the system's ability to help learners successfully regulate their learning process and provide them with formative or insightful information on their current state of learning. However, much of the research on ITS design still relies on traditional HCI evaluation methods used to identify issues related to usability, cognitive load, or preference. While this learning effectiveness measurement is reasonable, the effectiveness measure for an ITS in this context is one of learner mastery rather than system effectiveness.

Therefore, instead of a traditional approach, we adopted Keller's ARCS model of motivation [9] to plan a longitudinal user study to evaluate the learners' learning experience and development of their mastery of embouchure techniques. It is a well-known method to evaluate stimuli for learners' motivation and performance. Over the years, numerous researchers have applied its concept to design educational content with emerging technology, such as gamification, e-learning, and mobile learning, and demonstrated its effectiveness successfully [7, 8, 13], showing it can be used to influence learners' motivation positively in various educational contexts and learning goals.

The ARCS model categories are described as follows. "Attention" refers to the response of learners to perceive instructional stimuli. "Relevance" is about helping learners connect their prior learning experiences to the instruction provided. "Confidence" refers to emphasizing the importance of building positive expectations for the performance of students in the learning task. "Satisfaction" will be achieved at the end of the learning process when learners are allowed to practice newly acquired knowledge or skills. Based on the four distinctive categories of the ARCS model, we have developed both a questionnaire with a 5-point Likert scale for quantitative user studies and semi-structured interview questions to investigate novice oboe learners' learning experience qualitatively in future work.

5 CONCLUSION

We have illustrated the creation of an intelligent learning system for novice oboe learners. We have also proposed a framework to describe the three major processes involved in this system from a human-centered machine learning perspective. Implementing this interactive system, we successfully demonstrated how the framework can be applied to transform traditional, unsupervised oboe practice into a digitally enhanced learning experience using an SVM and interactive particle swarm animations. Moreover, we have discussed a set of guidelines and principles that can help researchers and designers develop intuitive, informative, and engaging intelligent learning systems in the music education context. While we showed the potential of our framework and system through a preliminary experiment, in the future, we will need to collect much more data from different participants across different contexts, build more robust models, and compare them in real-world environments to see if they can be sufficiently generalized to support various novice oboe learners' embouchure training issues under different situations. We expect that the impact of intelligent tutoring systems in this context will increase if future researchers put emphasis on (1) investigating the user's practice environments and any underlying mechanism that accelerates their learning process; (2) applying a deep learning approach to provide better classification and regression results; and (3) conveying simple yet informative visual content so the user can understand specific mistakes they make as well as for the user can proactively seek proper instructional methods or knowledge to correct it.

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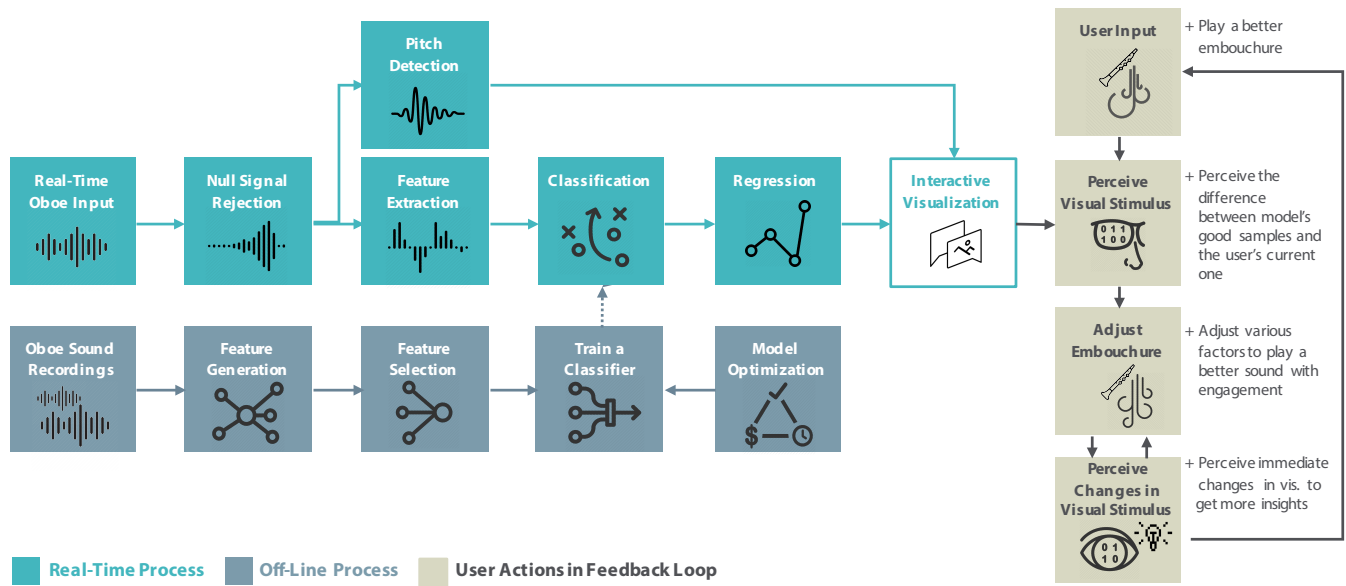


Figure 2: A proposed framework to use and augment the existing AI/ML process and understand how it can be applied to design interactive tutoring systems that are more learner-centered and interactive for real-time music practice and education

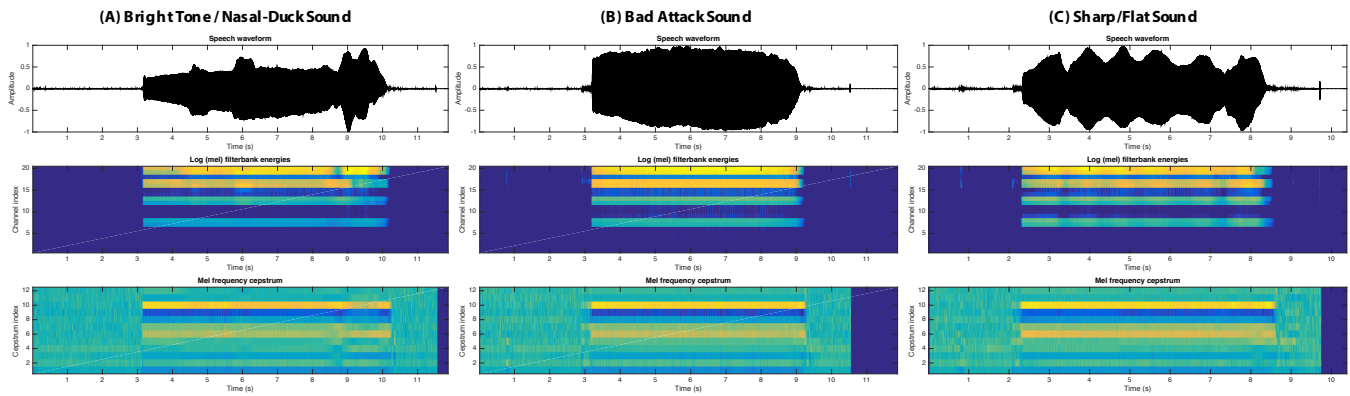


Figure 3: Computational representations of low-quality samples using mel filter-bank energy and MFCC. (Left) 'Nasal-duck' example (Middle) 'Bad Attack' (Right) 'Sharp/flat sound'

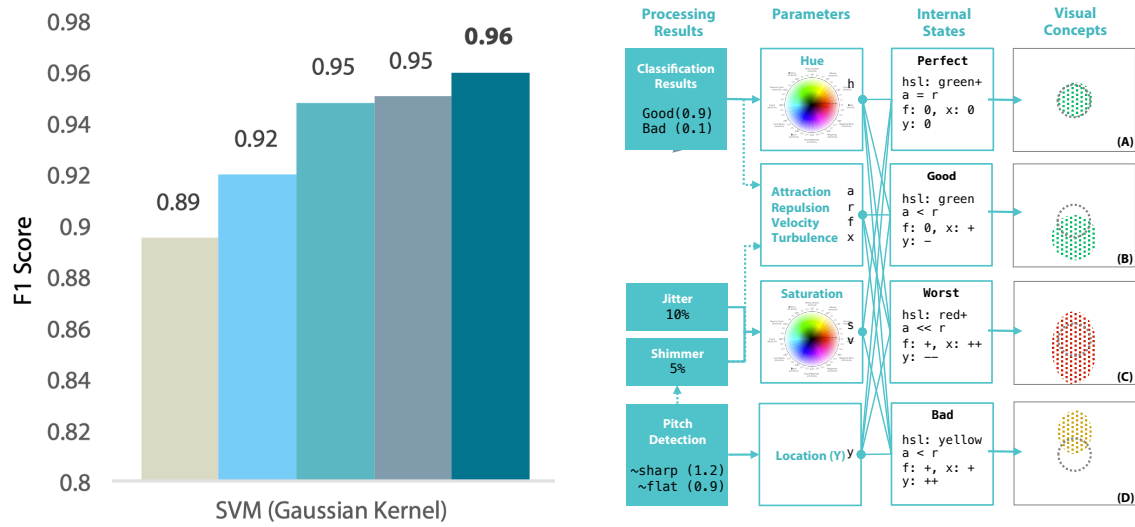


Figure 4: (Left) F1 score of SVM with a Gaussian kernel (Right) Visual mapping strategy between inference and visualization